



A comparative formulation of DKMQ, DSQ and MITC4 quadrilateral plate elements with new numerical results based on *s-norm* tests

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ABSTRACT

This paper presents a review of the formulation of four plate bending quadrilateral elements and new advanced validation test results. The four elements are DKQ, MITC4, DSQ and DKMQ published between 1982 and 1993. The first is valid for thin plates and the three others include transverse shear effects. All elements have 4 nodes and 3 *dof* per node (one displacement and two rotations) and give good results in classical benchmark and patch tests for thin plates. The paper presents the respective formulation of the four elements in a unified and comparative manner, starting from the formulation of the classical DKQ element published in 1982. The paper includes new convergence behavior analysis using the *s-norm* tests. The numerical results using DKMQ, DSQ and MITC4 elements show the optimal and uniform convergence for thin and thick plates with a slight advantage for DKMQ due to quadratic terms for the rotation variables.

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1. Introduction

The phenomenon of shear locking in plate bending finite elements based on Reissner-Mindlin theory [1,2] has been the subject of a large number of theses and papers during the last 45 years. To overcome this numerical phenomenon, reduced and selective integration [3–5] in displacement formulations have been used with incomplete success considering slenderness ratio, element distortion, individual element test, etc. Mixed formulation [6] and Assumed Natural Strain (ANS) proposed by Hughes and Tezduyar [7] and MacNeal [8] have been better alternatives to overcome the problem of shear locking. ANS has been found a very effective method used by many authors to develop new finite elements based on Reissner-Mindlin plate theory. One of the simplest and effective ANS formulation to obtain a 4 nodes, three degrees of freedom (*dof*) per node, plate bending element with Transverse Shear (TS) effects included is due to Bathe and Dvorkin [9,10] who proposed the effective nowadays well-known MITC4 [11] element (Mixed Interpolation of Tensorial Components).

Another approach developed by Batoz and former students was to include transverse shear effects in the element DKQ (Discrete Kirchhoff Quadrilateral). DKQ, published in 1982 [12], overcomes the C^1 continuity requirement when considering the Kirchhoff thin plate theory by using the Reissner-Mindlin theory but neglecting

TS energy and imposing discrete Kirchhoff constraints on edges (see also [13] for the triangular element DKT). Rotation components are based on incomplete quadratic polynomials. Extended validation based on individual element tests, patch tests, convergence tests have been performed and the DKQ element is included in the library of several softwares. Hence based on the DKQ element Batoz and Lardeur proposed the quadrilateral element called DSQ [14,15] (Discrete Shear Quadrilateral) still with 3 *dof* per node, where TS effects have been considered using element equilibrium equations and shear constitutive equations to define constant shear strains along the four edges of the element (see also [16,17] for the triangular version DST). DST and DSQ elements give overall good behavior for the analysis of isotropic and composites thin to thick plates but the transverse shear contribution is a bit complicated and patch tests for very thick plates were not fully satisfied.

Sharing several aspects of the formulation of DKQ and DSQ, Katili [18] proposed in 1993 the element DKMQ (Discrete Kirchhoff-Mindlin Quadrilateral) using simplified assumed transverse shear strains along the element sides (see also the formulation of the companion triangular element DKMT [19]). Both DKMT and DKMQ are valid for thin to thick plates, with good convergence properties and full satisfaction of the patch tests for constant curvature strain fields. Elements DST, DKMT, DSQ and DKMQ are free of shear locking by element constructions as they reduce respectively to DKT and DKQ for thin plates. Their difference relies in the representation of the transverse shear strain energy. The

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application of the DKMQ element in composite structures, stochastic and buckling analysis is presented in [20–22]. The extension of DKMQ plate element to a DKMQ24 shell element using Naghdi-Reissner-Mindlin shell theory taking into account warping effects and bending-membrane coupling is presented in [23–27].

A new development based on modified Timoshenko beam theory and modified Reissner-Mindlin plate theory has been proposed recently for beam and plate bending problems [28–32]. With this approach the shear locking problem does not occur due to the strong interdependence between the bending displacement and rotations.

Up to the middle of the 1990s the engineering community involved in finite element development technology was mainly emphasizing on individual element tests (proper rank for example), patch tests for constant stress situations, convergence analysis for particular displacements, (or less common stresses), at particular points of the structure with reference to a known analytical solution (or more often a numerical solution obtained by a fine mesh using reference-proved elements), for regular and distorted meshes. The shear locking phenomenon in plate finite elements based on various formulations and in particular mixed and assumed natural strains did attract the mathematical oriented community. As a result, new methods and tests to assess element behavior and convergence have been proposed during the last 25 years. Among others the so called *inf-sup* and *s-norm* tests [33–43].

The *inf-sup* condition for plate bending elements is related to the verification of the lack of shear locking. For the MITC4 element *inf-sup* tests have been performed to assess the monotonic and optimal convergence for thin plates [33–43]. For the DSQ and DKMQ elements shear locking cannot occur since those elements converge to DKQ for thin plates. However, a simple indicator for lack of shear locking has been used in this paper by computing the eigenvalues of the TS shear stiffness matrix of an individual element in an infinite mesh to confirm the lack of shear locking. The technique has been presented in [15,44,45].

The method called *s-norm* proposed in [33,34,38,39] is an appropriate method to evaluate the impact of the approximation fields and the element formulation on the whole structure. Numerical results based on the *s-norm* tests can evaluate the capacity for uniform and optimal convergence. Uniform convergence means that the convergence does not depend on the thickness of the problem under study while optimal convergence means that the rate of convergence and interpolations used in the formulation of the element are of the same order. The application of *s-norm* was extensively used in [40–43] to evaluate various MITC plate and shell elements.

The paper is organized as follows. Some aspects of the Reissner-Mindlin plate theory (kinematic, constitutive, equilibrium equations and modified *Hu-Washizu* functional) are recalled in Section 2. In Sections 3–6 the respective formulation of the elements DKQ, MITC, DSQ and DKMQ is presented using a unified notation. Order of presentation is simply the year of publication. Sections 7 and 8 deal with the pass evaluation tests and *s-norm* method. In particular, the convergence behavior of the four elements for various isotropic plate situations using the *s-norm* is presented in Section 9. Concluding remarks, acknowledgements and references are given at the end.

2. General aspects of the reissner-mindlin plate theory [1,2]

The Reissner-Mindlin (or first order RM) plate theory [1,2] is widely used for the finite element analysis of thin to thick isotropic and composite plates:

- Kinematic assumptions, displacement and strains:

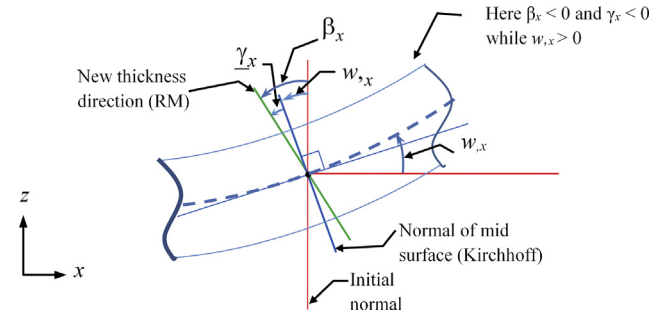


Fig. 1. Deformation of the normal vector with shear deformation at plane z - x .

Several assumptions are used in the linear first order RM plate theory (Fig. 1).

- A straight line normal to the mid-surface before deformation remains straight but not necessarily normal to the deformed mid-surface.
- The distance between points located on the straight line remains constant (no normal strain).

Displacement variables (u , v , w) in the Cartesian coordinates system x , y and z respectively are assumed to be independent in the following manner, z is the thickness direction.

$$u = z\beta_x(x, y), \quad v = z\beta_y(x, y), \quad w = w(x, y) \quad (1)$$

where β_x and β_y are the rotations in the z - x and z - y planes respectively. From (1) the linear bending strains and curvatures are given by:

$$\{\epsilon\} = z\{\chi\} \quad (2)$$

where

$$\{\epsilon\} = \langle \epsilon \rangle^T = \langle \epsilon_x \quad \epsilon_y \quad \epsilon_{xy} \rangle^T; \quad \{\chi\} = \langle \chi \rangle^T = \langle \chi_x \quad \chi_y \quad \chi_{xy} \rangle^T \quad (3)$$

The curvatures are given by:

$$\{\chi\} = \begin{Bmatrix} \chi_x \\ \chi_y \\ \chi_{xy} \end{Bmatrix} = \begin{Bmatrix} \beta_{x,x} \\ \beta_{y,y} \\ \beta_{x,y} + \beta_{y,x} \end{Bmatrix} \quad (4)$$

($\beta_{x,x}$ and $\beta_{y,y}$ denotes the first derivatives of β_x with respect to x and y respectively).

The transverse shear strains can be written as:

$$\{\gamma\} = \begin{Bmatrix} \gamma_x \\ \gamma_y \end{Bmatrix} = \begin{Bmatrix} \beta_x + w_{,x} \\ \beta_y + w_{,y} \end{Bmatrix} \quad (5)$$

- Generalized constitutive equations:

The constitutive equations for bending are given by:

$$\{M\} = \begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} = [H_b] \begin{Bmatrix} \chi_x \\ \chi_y \\ \chi_{xy} \end{Bmatrix} \quad (6)$$

where the bending moments are M_x , M_y and M_{xy} . For a linear elastic, isotropic homogeneous material, $[H_b]$ is defined as:

$$[H_b] = D_b \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu)/2 \end{bmatrix}; \quad D_b = \frac{Eh^3}{12(1-\nu^2)} \quad (7)$$

D_b is the bending plate rigidity, E the Young's modulus, h the thickness of the plate, and ν the Poisson's ratio.

The constitutive equation for the shear forces are given by:

$$\{T\} = \begin{Bmatrix} T_x \\ T_y \end{Bmatrix} = [H_s] \begin{Bmatrix} \gamma_x \\ \gamma_y \end{Bmatrix} \quad (8)$$

where the shear forces are T_x and T_y . For a linear elastic, isotropic homogeneous material, $[H_s]$ is simply:

$$[H_s] = D_s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad D_s = \kappa Gh \quad (9)$$

where κ is the shear correction factor (usually 5/6 for an isotropic homogeneous plate), the shear modulus is $G = E/2(1 + \nu)$.

– Equilibrium equations:

The internal forces can be used to define the equilibrium equations. Consider a plate subjected to a static distributed load f_z (Fig. 2) acting on the mid surface:

The equilibrium equations can be written as follows.

$$T_{x,x} + T_{y,y} + f_z = 0 \quad (10)$$

$$M_{x,x} + M_{xy,y} - T_x = 0 \quad (11)$$

$$M_{xy,x} + M_{y,y} - T_y = 0 \quad (12)$$

where T_x , T_y , M_x , M_y and M_{xy} are the shear forces and bending moments per unit length.

– Hu-Washizu principle:

Various variational principles can be proposed related to the RM plate theory. In our paper the modified Hu-Washizu principle is relevant for ongoing discussions. Consider the functional:

$$\Pi = \Pi_{int} - \Pi_{ext} \quad (13)$$

$$\Pi_{ext} = \int_A w f_z dA \quad (14)$$

$$\Pi_{int} = \Pi_{int}^b + \Pi_{int}^s \quad (15)$$

$$\Pi_{int}^b = \frac{1}{2} \int_A \langle \chi \rangle [H_b] \{ \chi \} dA \quad (\text{bending energy}) \quad (16)$$

$$\Pi_{int}^s = \frac{1}{2} \int_A \langle \gamma \rangle [H_s] \{ \gamma \} dA + \int_{A^e} \langle T \rangle (\{ \gamma \} - \{ \underline{\gamma} \}) ds \quad (\text{shear energy}) \quad (17)$$

Where

$\langle \gamma \rangle = \langle \gamma_x \quad \gamma_y \rangle$ are independent (assumed) strain variables and $\langle T \rangle = \langle T_x \quad T_y \rangle$ are independent shear forces.

The variation of Π with respect to the shear forces leads to the constraint equation:

$$\int_A \langle \delta T \rangle (\{ \gamma \} - \{ \underline{\gamma} \}) dA = 0 \quad (18)$$

At the finite element level, the above equation can be extended on any element side and expressed as:

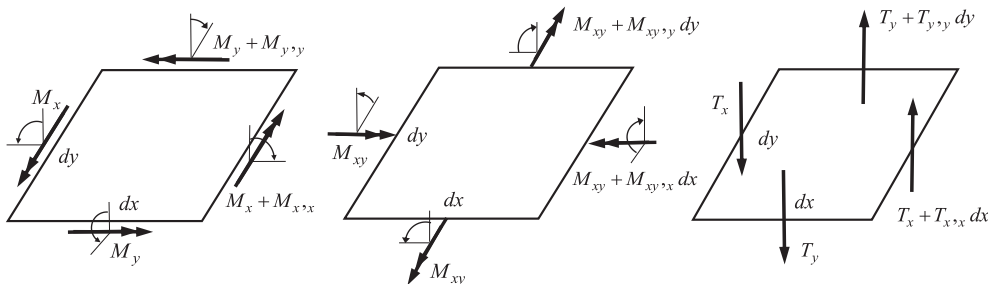


Fig. 2. Moments, shear forces and loads, on a $dx \times dy$ plate mid-surface element.

$$\int_0^{L_k} \delta T_s (\gamma_s - \underline{\gamma}_s) ds = 0 \quad (19)$$

where T_s and $\underline{\gamma}_s$ are the shear force and assumed shear strain along a side of length L_k . If the assumed shear force and strains are constant on the side, then we obtain:

$$\underline{\gamma}_{s_k} = \frac{1}{L_k} \int_0^{L_k} \gamma_s ds \quad (20)$$

3. Formulation of DKQ element [12]

The DKQ element was first published in 1982 [12]. DKQ element has 4 nodes with 3 degrees of freedoms each, which are: w (translation in the z direction), β_x (rotation in the x - z plane) and β_y (rotation in the y - z plane). Incomplete quadratic rotation fields for β_x and β_y are considered in terms of rotations at the four corners and of $\Delta\beta_{s_k}$ a temporary variable at mid-side i - j (Fig. 3).

On each side i - j , the normal rotation β_n is a linear function of s , while rotation β_s is quadratic in s (Fig. 4).

In a hierarchical form:

$$\beta_n = \left(1 - \frac{s}{L_k}\right) \beta_{n_i} + \frac{s}{L_k} \beta_{n_j} \quad (21)$$

$$\beta_s = \left(1 - \frac{s}{L_k}\right) \beta_{s_i} + \left(\frac{s}{L_k}\right) \beta_{s_j} + 4 \frac{s}{L_k} \left(1 - \frac{s}{L_k}\right) \Delta\beta_{s_k} \quad (22)$$

Based on (21) and (22), the rotation functions of β_x and β_y can be written in an element as:

$$\beta_x = \sum_{i=1}^4 N_i \beta_{x_i} + \sum_{k=5}^8 P_k C_k \Delta\beta_{s_k}; \quad \beta_y = \sum_{i=1}^4 N_i \beta_{y_i} + \sum_{k=5}^8 P_k S_k \Delta\beta_{s_k} \quad (23)$$

or

$$\begin{Bmatrix} \beta_x \\ \beta_y \end{Bmatrix} = [N_L] \{u_n\} + [N_Q] \{\Delta\beta_{s_n}\} \quad (24)$$

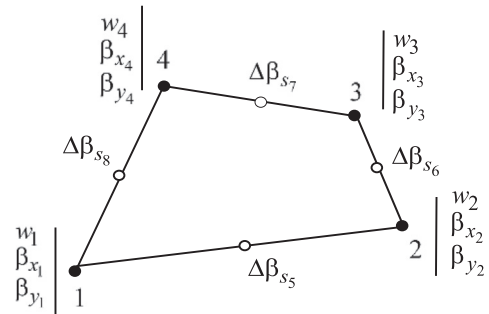


Fig. 3. Kinematic variable of corner node and temporary variable at mid side i - j .

isoparametric representation of both the coordinates x and y , and the independent displacement w and rotation fields β_x and β_y as follows:

$$w = \sum_{i=1}^4 N_i w_i; \quad \beta_x = \sum_{i=1}^4 N_i \beta_{x_i}; \quad \beta_y = \sum_{i=1}^4 N_i \beta_{y_i} \quad (38)$$

where N_i are bilinear shape functions as defined in Table 1. C^0 continuity is guaranteed. The relationship between curvature strains (4) and nodal variables is defined as follow:

$$\{\chi\} = [B_b] \{u_n\} \quad (39)$$

where

$$[B_b] = \begin{bmatrix} 0 & N_{i,x} & 0 & \dots & 0 \\ \dots & 0 & 0 & N_{i,y} & \dots \\ 0 & N_{i,y} & N_{i,x} & \dots & 0 \end{bmatrix} \quad i = 1, 2, 3, 4 \quad (40)$$

$$N_{i,x} = j_{11} N_{i,\xi} + j_{12} N_{i,\eta}; \quad N_{i,y} = j_{21} N_{i,\xi} + j_{22} N_{i,\eta} \quad (41)$$

$N_{i,x}$ and $N_{i,y}$ denote the first derivative of the shape function with respect to x and y respectively and j_{ij} the components of the inverse of the Jacobian matrix $[j]$. Matrix $[B_b]$ defined in (40) for MITC4 is the same as $[B_{b\theta}]$ defined in (26) for DKQ.

Independent assumed shear strains γ_x, γ_y are expressed in terms of natural shear strain components as follows:

$$\{\gamma\} = \begin{Bmatrix} \gamma_x(\xi, \eta) \\ \gamma_y(\xi, \eta) \end{Bmatrix} = \begin{bmatrix} j_{11} & j_{12} \\ j_{21} & j_{22} \end{bmatrix} \begin{Bmatrix} \gamma_\xi \\ \gamma_\eta \end{Bmatrix} = [j] \begin{Bmatrix} \gamma_\xi \\ \gamma_\eta \end{Bmatrix} \quad (42)$$

$$\begin{Bmatrix} \gamma_\xi(\eta) \\ \gamma_\eta(\xi) \end{Bmatrix} = [N_\gamma] \begin{Bmatrix} \gamma_{\xi n} \\ \gamma_{\eta n} \end{Bmatrix} \quad (43)$$

$$[N_\gamma] = \begin{bmatrix} \frac{1}{2}(1-\eta) & 0 & \frac{1}{2}(1+\eta) & 0 \\ 0 & \frac{1}{2}(1+\xi) & 0 & \frac{1}{2}(1-\xi) \end{bmatrix} \quad (44)$$

where

$$\langle \gamma_{\xi n} \rangle = \langle \gamma_{\xi 5} \quad \gamma_{\eta 6} \quad \gamma_{\xi 7} \quad \gamma_{\eta 8} \rangle$$

It can be seen from Eqs. (42)–(44) and (Fig. 6) that the natural shear strains γ_ξ is constant in the ξ direction and linear in the η direction. On the other hand γ_η is constant in the η direction and linear in the ξ direction. These two natural strains γ_ξ and γ_η have a similar variation in ξ and η as $w_{,\xi}$ and $w_{,\eta}$ so that the natural strains have a consistent approximation in terms of ξ and η .

The relationship between the shear strains in natural coordinate with the shear strains in the tangential coordinate system can simply be defined with the following relations:

$$\{\gamma_{\xi n}\} = [A_\gamma] \{\gamma_{s n}\} \quad (45)$$

where

$$\langle \gamma_{s n} \rangle = \langle \gamma_{s 5} \quad \gamma_{s 6} \quad \gamma_{s 7} \quad \gamma_{s 8} \rangle$$

and

$$[A_\gamma] = \begin{bmatrix} \frac{L_5}{2} & 0 & 0 & 0 \\ 0 & \frac{L_6}{2} & 0 & 0 \\ 0 & 0 & -\frac{L_7}{2} & 0 \\ 0 & 0 & 0 & -\frac{L_8}{2} \end{bmatrix} \quad (46)$$

From (42)–(46) we obtain:

$$\{\gamma\} = [j] [N_\gamma] [A_\gamma] \{\gamma_{s n}\} \quad (47)$$

On each side i - j , we recall that w , β_x and β_y have a linear variation in s (see Fig. 7):

$$\gamma_s = w_{,s} + \beta_s; \quad w = \left(1 - \frac{s}{L_k}\right) w_i + \frac{s}{L_k} w_j; \quad \beta_s = \left(1 - \frac{s}{L_k}\right) \beta_{s_i} + \frac{s}{L_k} \beta_{s_j} \quad (48)$$

Using Eqs. (48) and (20), we obtain:

$$\begin{aligned} \gamma_{s_k} &= \frac{1}{L_k} (w_j - w_i) + \frac{1}{2} \beta_{s_i} + \frac{1}{2} \beta_{s_j} \\ &= \frac{w_j - w_i}{L_k} + \frac{1}{2} (C_k \beta_{x_i} + S_k \beta_{y_i} + C_k \beta_{x_j} + S_k \beta_{y_j}) \end{aligned} \quad (49)$$

Using (49) for all sides we get:

$$\{\gamma_{s n}\} = [A_u] \{u_n\} \quad (50)$$

where $[A_u]$ is the same as (34). Introducing (50) into (47) we have

$$\{\gamma\} = [B_s] \{u_n\} \quad (51)$$

with

$$[B_s] = [j] [N_\gamma] [A_\gamma] [A_u] \quad (52)$$

Introducing (39)–(41) into (16), the bending energy of element MITC4 can then be written classically as follows:

$$\Pi_{int}^b = \frac{1}{2} \langle u_n \rangle [k_b] \{u_n\} \quad (53)$$

where the bending stiffness matrix is:

$$[k_b] = \int_A [B_b]^T [H_b] [B_b] dA \quad (54)$$

Introducing (51), (52) into (17)–(20), the transverse shear energy of an element is written as follows:

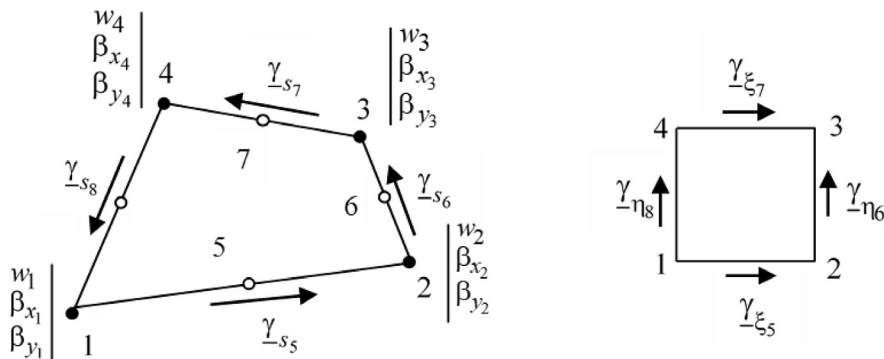


Fig. 6. Constant shear strains on each side of element in tangential and natural coordinate systems.

$$\Pi_{int}^s = \frac{1}{2} \langle u_n \rangle [k_s] \{u_n\} \quad (55)$$

where the shear stiffness matrix is:

$$[k_s] = \int_A [B_s]^T [H_s] [B_s] dA \quad (56)$$

The total stiffness of the MITC4 element is then:

$$[k] = [k_b] + [k_s] \quad (57)$$

As for DKQ a 2×2 Gauss integration scheme can be used for the numerical integration involved in (54) and (56).

5. Formulation of DSQ element [14,15]

The formulation of DSQ (Discrete Shear Quadrilateral) element was proposed by Batoz and Lardeur [14,15] in 1990 (written in french). Details on the companion triangular element DST have been published in english (see [16,17]). The aim of DSQ was to take into account transverse shear effects starting with the formulation of DKQ but without introduction of additional nodal variables. In this section, we briefly recall the formulation of the DSQ element.

The bending part formulation of DSQ is the same as for DKQ (12).

In the DSQ element the independent assumed shear strains γ_x and γ_y are defined using the element equilibrium equations (10)–(12) and the constitutive equations (6)–(9).

Consider (5), then (8), (11), (12) and (6) we obtain:

$$\{\gamma\} = \begin{Bmatrix} \gamma_x \\ \gamma_y \end{Bmatrix} = [H_s]^{-1} [\bar{H}_b] \begin{Bmatrix} \beta_{x,xx} + \beta_{x,yy} + \beta_{y,xy} \\ \beta_{x,yx} + \beta_{y,xx} + \beta_{y,yy} \end{Bmatrix} \quad (58a)$$

where β_x and β_y are given by (23) and (24). We obtain in terms of the nodal variables:

$$\{\gamma\} = \begin{Bmatrix} \gamma_x \\ \gamma_y \end{Bmatrix} = [B_{sp}] \{u_n\} + [B_{s\Delta\beta}] \{\Delta\beta_{sn}\} \quad (58b)$$

with

$$[B_{sp}(\xi, \eta)]_{2 \times 12} = [H_s]^{-1} [\bar{H}_b] [T_j(\xi, \eta)] [T_\beta] \quad (59)$$

$$[B_{s\Delta\beta}(\xi, \eta)]_{2 \times 4} = [H_s]^{-1} [\bar{H}_b] [T_j(\xi, \eta)] [T_{\Delta\beta}(\xi, \eta)] \quad (60)$$

$$[\bar{H}_b]_{2 \times 6} = \frac{D_b}{D_s} \begin{bmatrix} 1 & (1-\nu)/2 & 0 & 0 & 0 & (1+\nu)/2 \\ 0 & 0 & (1+\nu)/2 & (1-\nu)/2 & 1 & 0 \end{bmatrix} \quad (61)$$

$$[T_j(\xi, \eta)]_{6 \times 6} = \begin{bmatrix} [t_j] & [0] \\ [0] & [t_j] \end{bmatrix}; \quad [t_j] = \begin{bmatrix} j_{11}^2 & j_{12}^2 & 2j_{11}j_{12} \\ j_{21}^2 & j_{22}^2 & 2j_{21}j_{22} \\ j_{11}j_{21} & j_{12}j_{22} & j_{11}j_{22} + j_{12}j_{21} \end{bmatrix} \quad (62)$$

$$[T_\beta]_{6 \times 12} = \frac{1}{4} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & -1 \end{bmatrix} \quad (63)$$

$$[T_{\Delta\beta}(\xi, \eta)]_{6 \times 4} = \begin{bmatrix} -C_5(1-\eta) & 0 & -C_7(1+\eta) & 0 \\ 0 & -C_6(1+\xi) & 0 & -C_8(1-\xi) \\ C_5\xi & -C_6\eta & -C_7\xi & C_8\eta \\ -S_5(1-\eta) & 0 & -S_7(1+\eta) & 0 \\ 0 & -S_6(1+\xi) & 0 & -S_8(1-\xi) \\ S_5\xi & -S_6\eta & -S_7\xi & S_8\eta \end{bmatrix} \quad (64)$$

In the general case of a quadrilateral element, the assumed Cartesian shear strains γ_x and γ_y are not linear in ξ and η due to (62). Hence instead of using a mean value on a side as in (19) we simply consider the value of γ_x and γ_y at the mid node k of each side (collocation technique). Hence:

$$\gamma_k = \langle C_k \ S_k \rangle \begin{Bmatrix} \gamma_x \\ \gamma_y \end{Bmatrix} = \langle C_k \ S_k \rangle \left([B_{sp}(\xi_k, \eta_k)] \{u_n\} + [B_{s\Delta\beta}(\xi_k, \eta_k)] \{\Delta\beta_{sn}\} \right) \quad (65)$$

Now as for DKQ and MITC4, Eq. (20) is used. Considering also (30), (31), and (65) we obtain on each side:

$$\frac{w_j - w_i}{L_k} + \frac{1}{2} (C_k \beta_{xi} + S_k \beta_{yi} + C_k \beta_{xj} + S_k \beta_{yj}) + \frac{2}{3} \Delta\beta_{sk} - \langle C_k S_k \rangle \left([B_{sp}(\xi_k, \eta_k)] \{u_n\} + [B_{s\Delta\beta}(\xi_k, \eta_k)] \{\Delta\beta_{sn}\} \right) = 0 \quad (66)$$

The first part of (66) is the same as for DKQ (involving $\Delta\beta_{sk}$), the second part results from the assumed TS strains defined at each mid node considering the equilibrium and constitutive equations on the plate. We see that the second part of (66) implies the four components $\Delta\beta_{sk}$ ($k = 5, 6, 7, 8$). Considering (66) at each of the four sides we can define the following matrix equation:

$$\{\Delta\beta_{sn}\} = [A_\Delta]_{DSQ}^{-1} [A_{uu}] \{u_n\} \quad (67)$$

where

$$[A_\Delta]_{DSQ} = -\frac{2}{3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} \langle C_5 \ S_5 \rangle [B_{s\Delta\beta}(\xi_5 = 0, \eta_5 = -1)] \\ \langle C_6 \ S_6 \rangle [B_{s\Delta\beta}(\xi_6 = +1, \eta_6 = 0)] \\ \langle C_7 \ S_7 \rangle [B_{s\Delta\beta}(\xi_7 = 0, \eta_7 = +1)] \\ \langle C_8 \ S_8 \rangle [B_{s\Delta\beta}(\xi_8 = -1, \eta_8 = 0)] \end{bmatrix} \quad (68)$$

$$[A_{uu}] = [A_u]_{4 \times 12} = \begin{bmatrix} \langle C_5 \ S_5 \rangle [B_{sp}(\xi_5 = 0, \eta_5 = -1)] \\ \langle C_6 \ S_6 \rangle [B_{sp}(\xi_6 = +1, \eta_6 = 0)] \\ \langle C_7 \ S_7 \rangle [B_{sp}(\xi_7 = 0, \eta_7 = +1)] \\ \langle C_8 \ S_8 \rangle [B_{sp}(\xi_8 = -1, \eta_8 = 0)] \end{bmatrix} \quad (69)$$

where $[A_u]$ is the same as (34).

Introducing Eq. (67) into Eq. (25) leads to:

$$\{\chi\} = [B_b] \{u_n\} \quad \text{with} \quad [B_b] = [B_{bb}] + [B_{b\Delta\beta}] [A_\Delta]_{DSQ}^{-1} [A_{uu}] \quad (70)$$

Introducing Eq. (67) into Eq. (58), we obtain:

$$\{\gamma\} = [B_s] \{u_n\} \quad \text{with} \quad [B_s] = [B_{sp}] + [B_{s\Delta\beta}] [A_\Delta]_{DSQ}^{-1} [A_{uu}] \quad (71)$$

The stiffness matrix for bending and shear are then defined using (53)–(57). Again numerical integration can be performed using 2×2 Gauss points.

As a result of (66) and (67) we can see that the elimination of $\Delta\beta_{sk}$ will destroy the C^0 continuity, however the proper rank is kept and DSQ will converge to DKQ for thin plates. Extensive numerical results have been published in [14].

6. Formulation of DKMQ element [18]

The DKMQ (Discrete Kirchhoff Mindlin Quadrilateral) element proposed by Katili [18] in 1993 combines some ideas and formulation aspects found in DKQ, MITC4 and DSQ to achieve a rather simple and efficient element valid for thin to thick plates. DKMQ has again 4 nodes with 3 degrees of freedom at each node (Fig. 3). The bending formulation is the same as for DKQ,

(see Eqs. (21)–(28)) and the assumed TS strain field is similar compared to MITC4, (see Eqs. (42)–(47)). In the spirit of DSQ, the independent transverse shear strains are using local equilibrium and constitutive equations considering each side as a beam in order to keep the C^0 continuity.

In [18–19], Katili proposed the assumed independent transverse shear strain γ_s along the side i - j , can be expressed as:

$$\gamma_s = \frac{D_b}{D_s} \beta_{s,ss} \quad (72)$$

with β_s previously defined in (22), on side i - j , (72) can be written as:

$$\gamma_{sk} = -\frac{2}{3} \phi_k \Delta \beta_{sk} \quad (73)$$

where the coefficient ϕ_k can be obtained for isotropic material [18,19]

$$\phi_k = \frac{D_b}{D_s} \frac{12}{L_k^2} = \frac{2}{\kappa(1-\nu)} \left(\frac{h}{L_k} \right)^2 \quad (74)$$

Where κ is the shear correction factor (usually $\kappa = 5/6$). The factor ϕ_k , which is characterizing the influence of shear, maintains the consistency of proposed element and precisely explains why DKMQ element behaves as either *Reissner-Mindlin* theory for thick plate or as *Kirchhoff-Love* theory for thin plate. In the thin plate problems, where factor ϕ_k is close to zero, shear deformation is automatically reduced. Accordingly, as the main positive result, the *shear locking* is automatically resolved by this Discrete Kirchhoff Mindlin method. If (73) is applied on all sides of the element, the following matrix relation is obtained:

$$\{\gamma_{sn}\} = [A_\phi] \{\Delta \beta_{sn}\} \quad (75)$$

$$[A_\phi] = -\frac{2}{3} \begin{bmatrix} \phi_5 & 0 & 0 & 0 \\ 0 & \phi_6 & 0 & 0 \\ 0 & 0 & \phi_7 & 0 \\ 0 & 0 & 0 & \phi_8 \end{bmatrix} \quad (76)$$

Introducing (75) into (47), the shear strains for DKMQ can be expressed as:

$$\{\gamma\} = \left\{ \begin{matrix} \gamma_x \\ \gamma_y \end{matrix} \right\} = [j] [N_\gamma] [A_\gamma] [A_\phi] \{\Delta \beta_{sn}\} \quad (77)$$

Considering Eqs. (20), (73) and (30) we obtain on each side:

$$\frac{w_j - w_i}{L_k} + \frac{1}{2} (C_k \beta_{xi} + S_k \beta_{yi} + C_k \beta_{xj} + S_k \beta_{yj}) + \frac{2}{3} (1 + \phi_k) \Delta \beta_{sk} = 0 \quad (78)$$

leading to

$$\{\Delta \beta_{sn}\} = [A_\Delta]_{DKMQ}^{-1} [A_u] \{u_n\} \quad (79)$$

with

$$[A_\Delta]_{DKMQ}^{-1} = -\frac{3}{2} \begin{bmatrix} \frac{1}{(1+\phi_5)} & 0 & 0 & 0 \\ 0 & \frac{1}{(1+\phi_6)} & 0 & 0 \\ 0 & 0 & \frac{1}{(1+\phi_7)} & 0 \\ 0 & 0 & 0 & \frac{1}{(1+\phi_8)} \end{bmatrix} \quad (80)$$

and $[A_u]$ given by (34).

Introduction of (79) into (25) we define the bending curvatures as:

$$\{\chi\} = [B_b] \{u_n\} \quad \text{where} \quad [B_b] = [B_{b\beta}] + [B_{b\Delta\beta}] [A_\Delta]_{DKMQ}^{-1} [A_u] \quad (81)$$

We introducing (79) into (77) obtain:

$$\{\gamma\} = [B_s] \{u_n\} \quad \text{where} \quad [B_s] = [j] [N_\gamma] [A_\gamma] [A_\phi] [A_\Delta]_{DKMQ}^{-1} [A_u] \quad (82)$$

where

$$[A_\phi] = [A_\phi] [A_\Delta]_{DKMQ}^{-1} = \begin{bmatrix} \frac{\phi_5}{(1+\phi_5)} & 0 & 0 & 0 \\ 0 & \frac{\phi_6}{(1+\phi_6)} & 0 & 0 \\ 0 & 0 & \frac{\phi_7}{(1+\phi_7)} & 0 \\ 0 & 0 & 0 & \frac{\phi_8}{(1+\phi_8)} \end{bmatrix} \quad (83)$$

The stiffness matrix for bending and shear using (53)–(57) is obtained by using 2×2 Gauss points.

Remarks:

- When ϕ_k , very small i.e. if $h \ll L_k$, then $[A_\Delta]_{DKMQ}^{-1} \approx [A_\Delta]_{DKQ}^{-1}$ and $[A_\phi] \approx [0]$ (transverse shear effects vanish). Therefore $DKMQ \approx DKQ$. As a consequence, DKMQ is free of shear locking.
- When ϕ_k , very large i.e. the element size $L_k \ll h$, then $[A_\Delta]_{DKMQ}^{-1} \approx [0]$ leading to $[B_b] \approx [B_{b\beta}]$ and $[A_\phi] \approx [I]$ leading to $[B_s] = [j] [N_\gamma] [A_\gamma] [A_u]$. As consequence for ϕ_k , very large, $DKMQ \approx MITC4$.

7. Past evaluation tests

The proper rank of the stiffness matrix of a single element (DKQ, MITC4, DSQ, DKMQ) for any aspect ratio and distortion has been verified in previous publications on the four elements. They satisfy an important *ellipticity condition*: only three zero eigenvalues representing the rigid body modes with the other nine all positive.

Also constant curvatures patch test are satisfied for large slenderness ratio (say $L/h \geq 100$) in accordance with the necessary element *consistency condition*.

All four elements are free of shear locking: DKQ by direct element formulation, DSQ and DKMQ because they converge to DKQ for thin plates. MITC4 is also free of shear locking by choosing consistent approximations to represent the shear strains in natural coordinates. Numerical tests can be used to show the lack of shear locking: convergence analysis for thin plates for different shapes, different boundary conditions, different loadings. Results can be found in the respective references of the elements.

A simple indicator to evaluate the possibility of shear locking is to compute the rank of the shear stiffness matrix of a single element in an assumed infinite mesh (Fig. 8).

For the three elements DKMQ, DSQ and MITC4 including transverse shear, we found that:

$$C_b = \frac{3}{\text{rank}[k'_s]} = \frac{3}{2} \quad (84)$$

The numerator (=3) of C_b represents the number of *dof* at node 1 of the single element. $[k'_s]$ is a 3×3 reduction of $[k_s]$ (12×12), considering full clamping at nodes 2, 3, 4. $C_b = 3/2$ has been found for the 3 elements and for any position of node 1 (distortion effect). From the literature, see [15,44,45] among others, a good indicator for the lack of shear locking is $C_b = 3/2$.

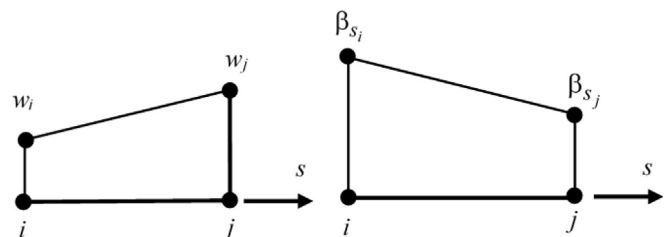


Fig. 7. Displacement w and rotation β_s on each side i - j .

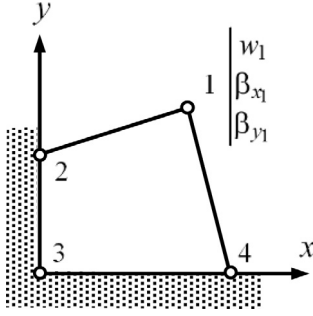


Fig. 8. A single element in an assumed infinite mesh.

A more mathematically oriented numerical test proposed for mixed, assumed strain based, (or any other formulation) to evaluate the risk of shear locking is the *inf-sup* test. MITC4 has been evaluated using that test (see [33–43]). The programming of this test is not easy and we did not consider necessary to perform the *inf-sup* tests for DSQ and DKMQ, relying on the inherent formulation of both elements.

In Table 2 we propose an evaluation of the characteristics and performances of the 4 elements based on our experience of programming and on results of classical tests (see references [12] for DKQ, [14] for DSQ, [18,20,22,23] for DKMQ, [9–11,34] for MITC4, and on the results reported in Section 9 of the present paper).

8. s-norm method

Uniform and optimal convergence of elements can be assessed using *s-norm* tests as defined in [38–40,46].

The relative error is computed using the general expression:

$$E_s = \frac{\|e^{FE}\|^2}{\|\Pi_{int}^{Ref}\|^2} = \frac{\|\Pi_{int}^{Ref} - \Pi_{int}^{FE}\|^2}{\|\Pi_{int}^{Ref}\|^2} \quad (85)$$

where

- $\|\Pi_{int}^{Ref}\|^2$ refers to the strain energy of the reference solution which can be defined by an analytical expression (like for circular plates, see [15,19,20]) or numerically by using a very fine mesh with “good” elements.
- Even if an analytical solution is available it is not always easy to express $\|\Pi_{int}^{Ref}\|^2$ and usually a very fine mesh is used. In the following section we consider meshes of 128×128 elements using DKMQ elements.

Table 2

Element characteristics based on classical evaluation tests and formulation aspects.

Element characteristics	DKQ	MITC4	DSQ	DKMQ
Transverse of shear strains are considered [11,14,18]	No	Yes	Yes	Yes
Pass the patch test for thin plates [15,18]	Yes	Yes	Yes	Yes
Pass the patch test for thick plates [11,18]	–	Yes	No	Yes
Relatively insensitive to distortion [11,12,18]	Yes	Yes	Yes	Yes
No shear locking for thin plates [12,15,34]	Yes	Yes	Yes	Yes
Simplicity of formulation and programming [11,12,18]	Yes	Yes	No	Yes
Proper rank (Exhibit no spurious zero-energy modes) [12,15,18]	Yes	Yes	Yes	Yes
Good convergence properties in classical engineering tests [11,12,14,15,18]	Yes	Yes	Yes	Yes
Possibility to obtain the exact stiffness matrix for a beam [14,15,29]	Yes	No	Yes	Yes

– For the present plate bending problems, we then have:

$$\|\Pi_{int}^{Ref}\|^2 = \sum_1^{NELT} \|\Pi_{int}^{Ref}\|_e^2 \quad (86)$$

NELT is the total number of elements in the reference mesh.

$$\|\Pi_{int}^{Ref}\|_e^2 = \frac{1}{2} \int_A \langle M_{Ref} \rangle [H_b]^{-1} \{M_{Ref}\} dA + \frac{1}{2} \times \int_A \langle T_{Ref} \rangle [H_s]^{-1} \{T_{Ref}\} dA \quad (87)$$

and

$$\|\Pi_{int}^{Ref} - \Pi_{int}^{FE}\|^2 = \sum_1^{NELT} \|\Pi_{int}^{Ref} - \Pi_{int}^{FE}\|_e^2 \quad (88)$$

$$\|\Pi_{int}^{Ref} - \Pi_{int}^{FE}\|_e^2 = \frac{1}{2} \int_A \langle \Delta M \rangle [H_b]^{-1} \{\Delta M\} dA + \frac{1}{2} \times \int_A \langle \Delta T \rangle [H_s]^{-1} \{\Delta T\} dA \quad (89)$$

where

$$\langle \Delta M \rangle = \langle M_{Ref} \rangle - \langle M_{FE} \rangle; \quad \langle \Delta T \rangle = \langle T_{Ref} \rangle - \langle T_{FE} \rangle \quad (90)$$

We integrate over the element area using (2×2) Gauss points in the reference mesh, so that (since the weighting coefficients of the 2×2 Gauss scheme are equal to one):

$$\|\Pi_{int}^{Ref}\|_e^2 = \frac{1}{2} \sum_{l=1}^{NPG} \langle M_{Ref}(\xi_l, \eta_l) \rangle [H_b]^{-1} \{M_{Ref}(\xi_l, \eta_l)\} J(\xi_l, \eta_l) + \frac{1}{2} \sum_{l=1}^{NPG} \langle T_{Ref}(\xi_l, \eta_l) \rangle [H_s]^{-1} \{T_{Ref}(\xi_l, \eta_l)\} J(\xi_l, \eta_l) \quad (91)$$

and

$$\|\Pi_{int}^{Ref} - \Pi_{int}^{FE}\|_e^2 = \frac{1}{2} \sum_{l=1}^{NPG} \langle \Delta M(\xi_l, \eta_l) \rangle [H_b]^{-1} \{\Delta M(\xi_l, \eta_l)\} J(\xi_l, \eta_l) + \frac{1}{2} \sum_{l=1}^{NPG} \langle \Delta T(\xi_l, \eta_l) \rangle [H_s]^{-1} \{\Delta T(\xi_l, \eta_l)\} J(\xi_l, \eta_l) \quad (92)$$

NPG is the total number of Gauss points in each element of the reference mesh.

And

$$\langle \Delta M(\xi_l, \eta_l) \rangle = \langle M_{Ref}(\xi_l, \eta_l) \rangle - \langle M_{FE}(\xi_l, \eta_l) \rangle \quad (93)$$

$$\langle \Delta T(\xi_l, \eta_l) \rangle = \langle T_{Ref}(\xi_l, \eta_l) \rangle - \langle T_{FE}(\xi_l, \eta_l) \rangle$$

$\langle M_{FE}(\xi_l, \eta_l) \rangle$ and $\langle T_{FE}(\xi_l, \eta_l) \rangle$ are the bending moments and shear forces in the coarse mesh, but one of the difficulty is to calculate the values of $\langle M_{FE}(\xi_l, \eta_l) \rangle$ and $\langle T_{FE}(\xi_l, \eta_l) \rangle$ at Gauss points (93) of each element in the reference mesh (with ξ_l and $\eta_l = \pm 1/\sqrt{3}$).

The reference solutions Π_{int}^{Ref} can either be obtained through analytical methods or numerical methods. In practice, analytical solutions can be obtained only when very simple geometries are considered. For these reasons, numerical methods are frequently preferred in obtaining reference solutions. When the exact solution is known, Eq. (85) become:

$$E_R = \frac{\|e^{FE}\|^2}{\|\Pi_{int}^{Ex}\|^2} = \frac{\|\Pi_{int}^{Ex} - \Pi_{int}^{FE}\|^2}{\|\Pi_{int}^{Ex}\|^2} \quad (94)$$

which states the orthogonality of the exact solution with respect to the finite element solution.

Π_{int}^{Ex} refers to the exact solution and Π_{int}^{FE} refers to the finite element model for a given mesh.

$$\|\Pi_{int}^{FE}\|^2 = \sum_1^{NELT} \|\Pi_{int}^{FE}\|_e^2 \quad (95)$$

NELT is the total number of elements in the reference mesh.

$$\|\Pi_{int}^{FE}\|_e^2 = \frac{1}{2} \int_A \langle M_{FE} \rangle [H_b]^{-1} \{M_{FE}\} dA + \frac{1}{2} \int_A \langle T_{FE} \rangle [H_s]^{-1} \{T_{FE}\} dA \quad (96)$$

In the *s-norm* method, the errors are represented by plotting curves Log of E_s in term of Log L_k , where L_k represents an indicator of the element size in a mesh (in our work L_k is the element size for square elements and an average element size for distorted meshes, like for circular plates). The resulting curves based on 2 values of L/h are compared with the slopes $p=1$ of the theoretical convergence (energy) formula:

$$E_s \cong cL_k^{2p} \quad (97)$$

where c is a constant.

In elements DSQ and DKMQ incomplete quadratic interpolations for the rotations are considered whereas for MITC4 linear approximations are considered. It is therefore important to check the three elements (for $L/h = 2$ and 10^4), with respect to $p = 1$.

9. Numerical results based on s-norm

The three elements DKMQ, DSQ and MITC4 have been programmed by the authors with results reported below for regular and distorted meshes, for rectangular and circular plates under uniform loading and for various slenderness ratio (from thick to very thin plates). For uniform loading, bilinear approximation for the transverse displacement $w(\xi, \eta)$ is used for all elements to define the equivalent load vectors. Regarding MITC4, our results are in agreement with those published in [33], obtained for thin

plates. Since DKMQ and DSQ converge to DKQ for thin plates, it is not necessary to report the results for DKQ. We only report the values for $L/h = 2$ and $L/h = 10^4$ to obtain the convergence slope.

9.1. Clamped square plate

We analyze a clamped square plate under uniform loading f_z where the details can be seen on Fig. 9. Because of symmetry, only area ABCD is considered with hard support conditions $w = \beta_y = \beta_x = 0$ along AB and AD. The symmetry condition $\beta_x = 0$ along BC and $\beta_y = 0$ along DC are taken into account.

Table 3 shows the percentage of bending and shear energy for a clamped square plate using $N \times N = 128 \times 128$ DKMQ elements for different L/h .

For a thick plate ($L/h = 2$), the shear energy reaches 61% but still bending exists since the bending energy is 39% of the total energy. Transverse shear effects quickly vanish for $L/h > 10^2$. In Fig. 10 we report the results obtained using the three elements in the extreme situations of $L/h = 2$ and $L/h = 10^4$, in a classical manner, i.e. E_s versus N number of elements between A and B.

For coarse mesh ($N = 4$) the results are different with an advantage to the family of DKQ (for the thin plate), DSQ and DKMQ, compared to MITC4, but all elements converge quickly and the relative errors E_s is very small ($< 0.1\%$). It is also observed that the same results are obtained for DSQ and DKMQ for thin and thick plates.

By using the solution of DKMQ elements with $N \times N = 128 \times 128$ as reference solution and considering two values of L/h ($L/h = 2$ and 10^4), Fig. 11 shows the convergence behavior of DKMQ, DSQ and MITC4 elements for $N = 4, 8, 16$ and 32 .

We confirm that DKMQ and DSQ have the same behavior as DKQ for $L/h > 10^2$. It is also observing that the convergence is uniform and linear for all three elements. Errors are a bit higher for MITC4 compared with the other elements. Results of MITC4 are in agreement with results presented in [34] and also recently [46] using MITC9 as the reference solution [41].

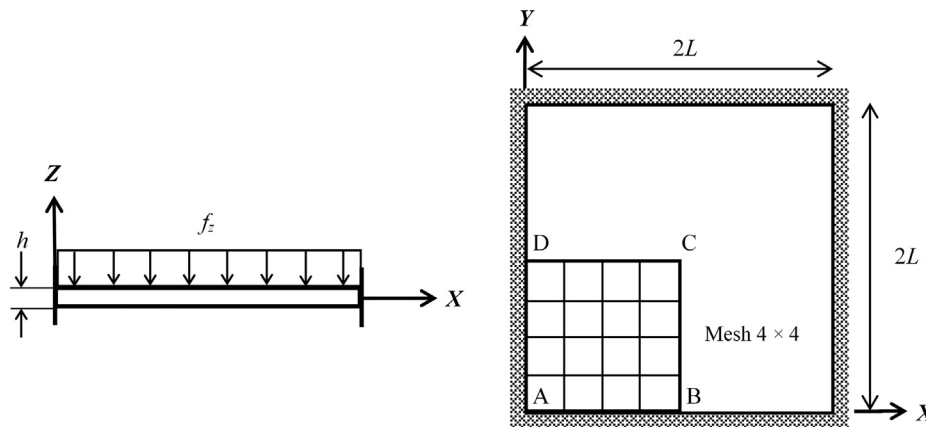


Fig. 9. Clamped square plate ($E = 1.7472 \times 10^7$; $\nu = 0.3$; $L = 1$; $f_z = -1$).

Table 3
Bending and shear energy for clamped square plate using $N \times N = 128 \times 128$ DKMQ elements for different L/h .

L/h	$\ \Pi_{int}^b\ ^2$ (%)		$\ \Pi_{int}^s\ ^2$ (%)		$\ \Pi_{int}^{Ref}\ ^2$ (%)	
2	3.266×10^{-8}	39.05%	5.097×10^{-8}	60.95%	8.363×10^{-8}	100%
4	3.525×10^{-7}	70.37%	1.063×10^{-7}	29.63%	3.589×10^{-7}	100%
10	3.896×10^{-6}	93.24%	2.823×10^{-7}	6.76%	4.178×10^{-6}	100%
10^2	3.891×10^{-3}	99.93%	2.791×10^{-6}	0.07%	3.894×10^{-3}	100%
10^3	3.891×10^0	100%	2.209×10^{-5}	0%	3.891×10^0	100%
10^4	3.891×10^3	100%	2.181×10^{-4}	0%	3.891×10^3	100%

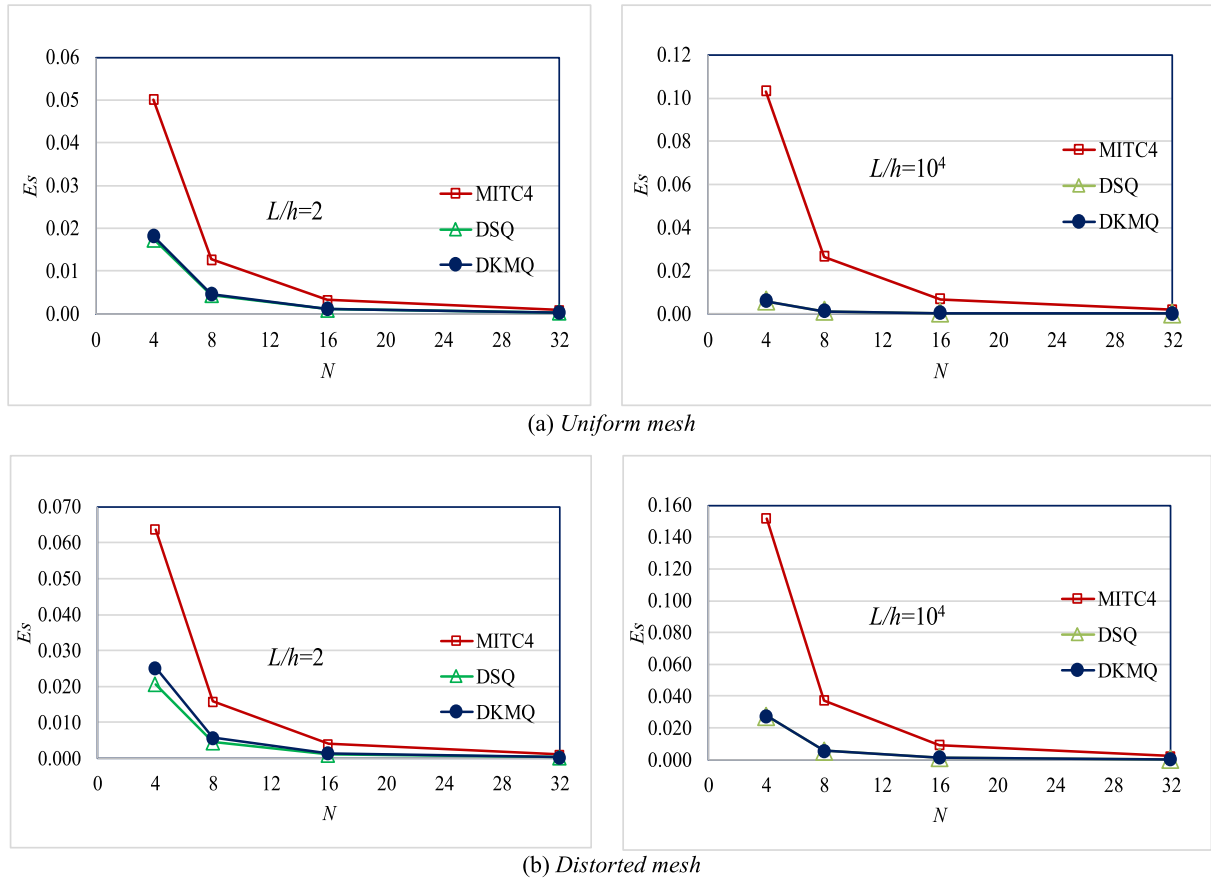


Fig. 10. Relative error for clamped square plate with uniform and distorted mesh for $L/h = 2$ and 10^4 .

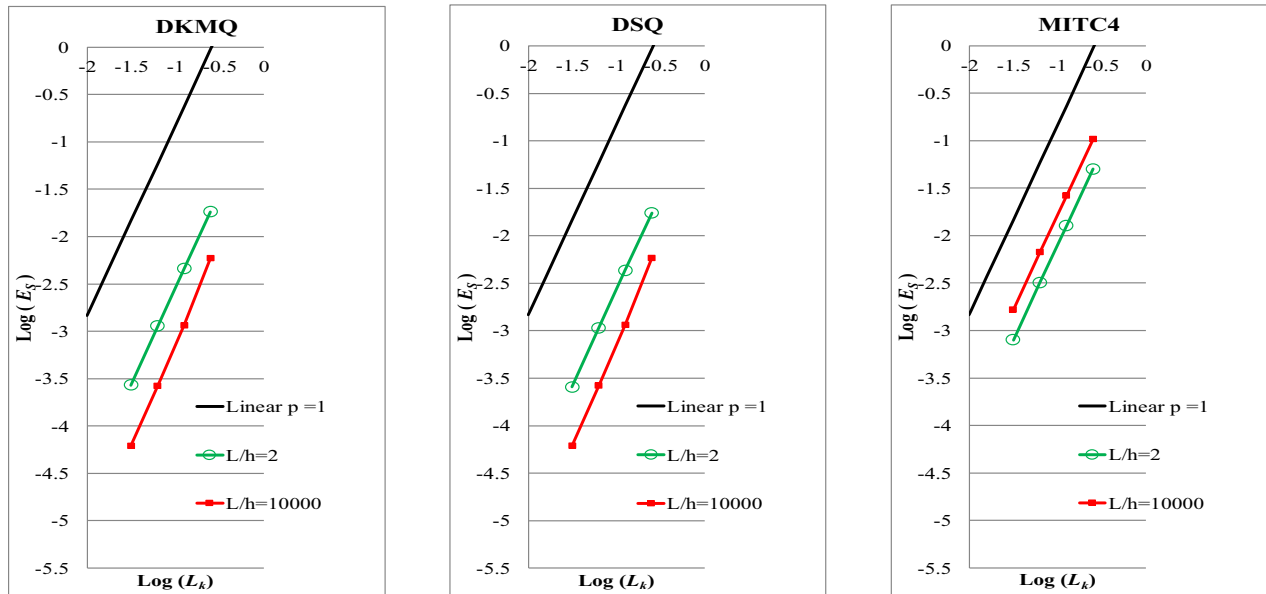


Fig. 11. Convergence behavior for clamped square plate by using s -norm with uniform mesh.

We also consider the effect of distortion (Fig. 12). The distorted mesh characteristics are as follows: each length $2L/3$ and $L/3$ on a side is successively divided by 2, 4, 8, 16 equal segments.

Fig. 13 shows the convergence behavior of the DKMQ, DSQ and MITC4 elements with the distorted mesh presented in Fig. 12 for $N = 4, 8, 16$ and 32 .

Similar comments for the distorted meshes can be made compared to uniform meshes (Fig. 11). Distortion of elements is almost insignificant for this test. Same results for MITC4 are reported in [46]. We can observe in Fig. 11 and 13 that the convergence behavior of MITC4 element in very thin plate ($L/h = 10^4$) is not as good as its behavior in very thick plate ($L/h = 2$).

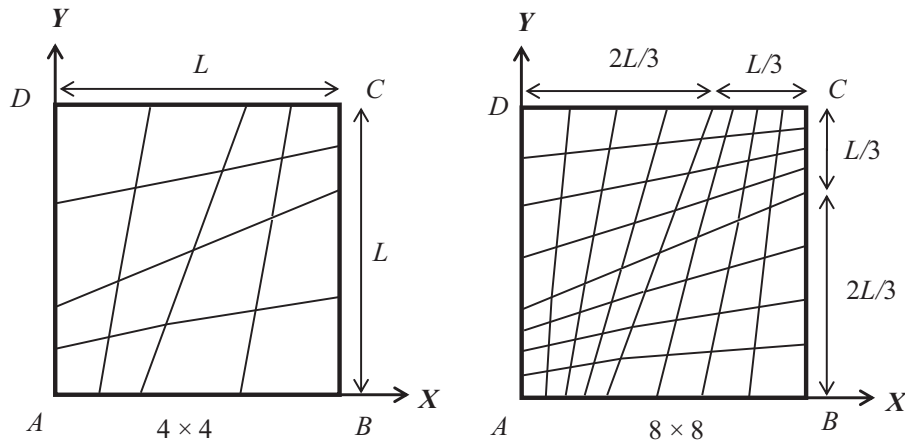


Fig. 12. Distorted mesh of square plate.

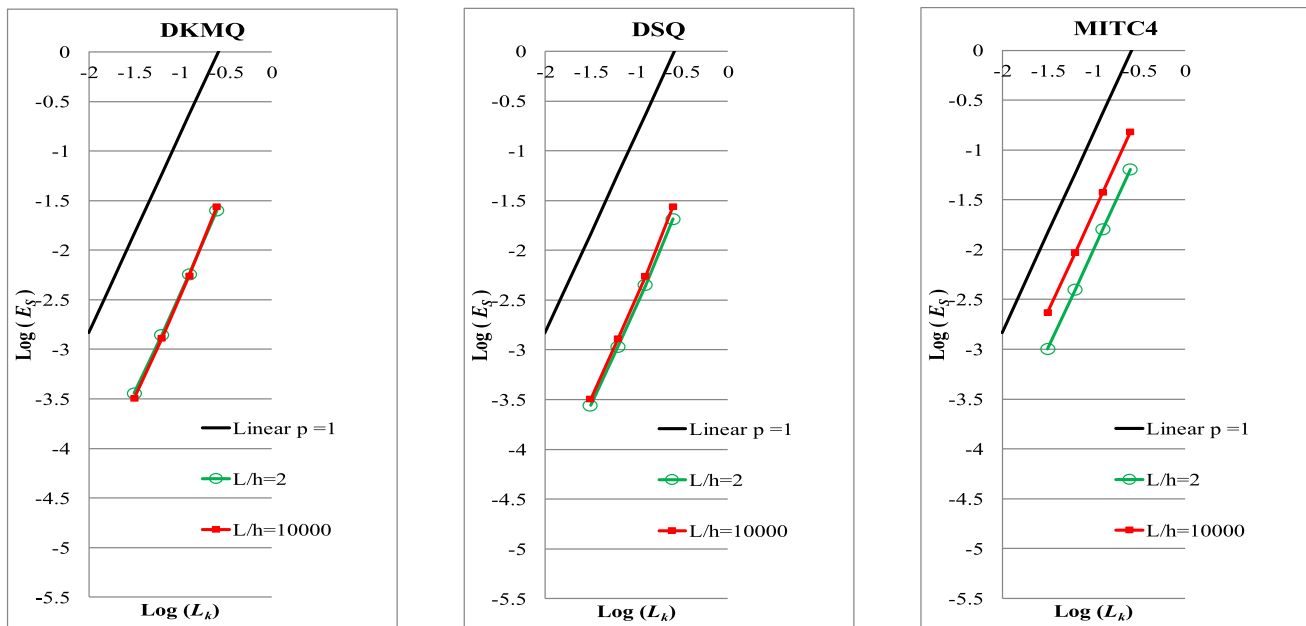
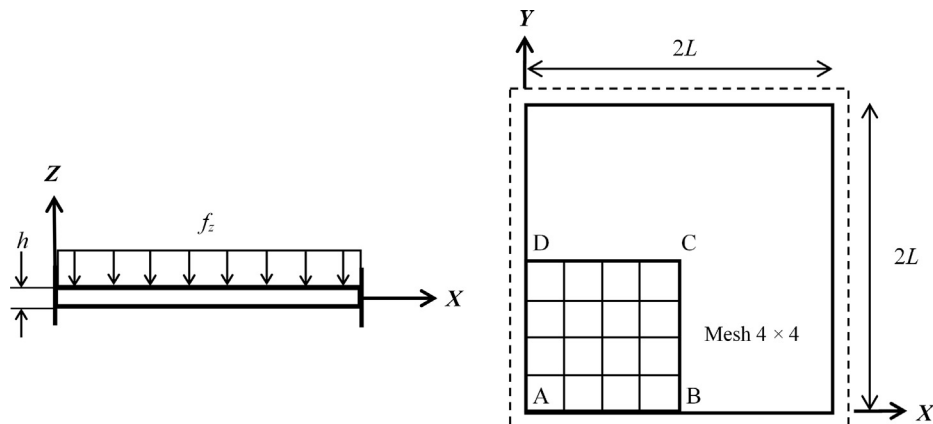
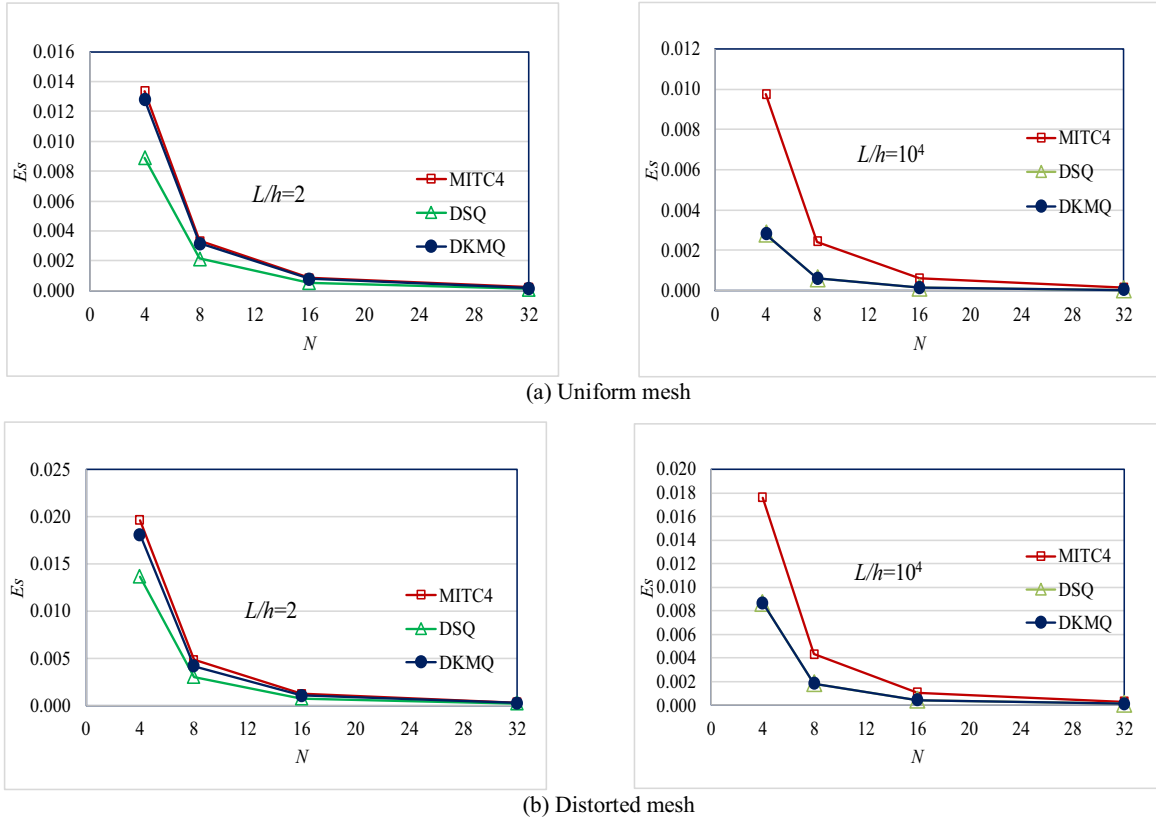
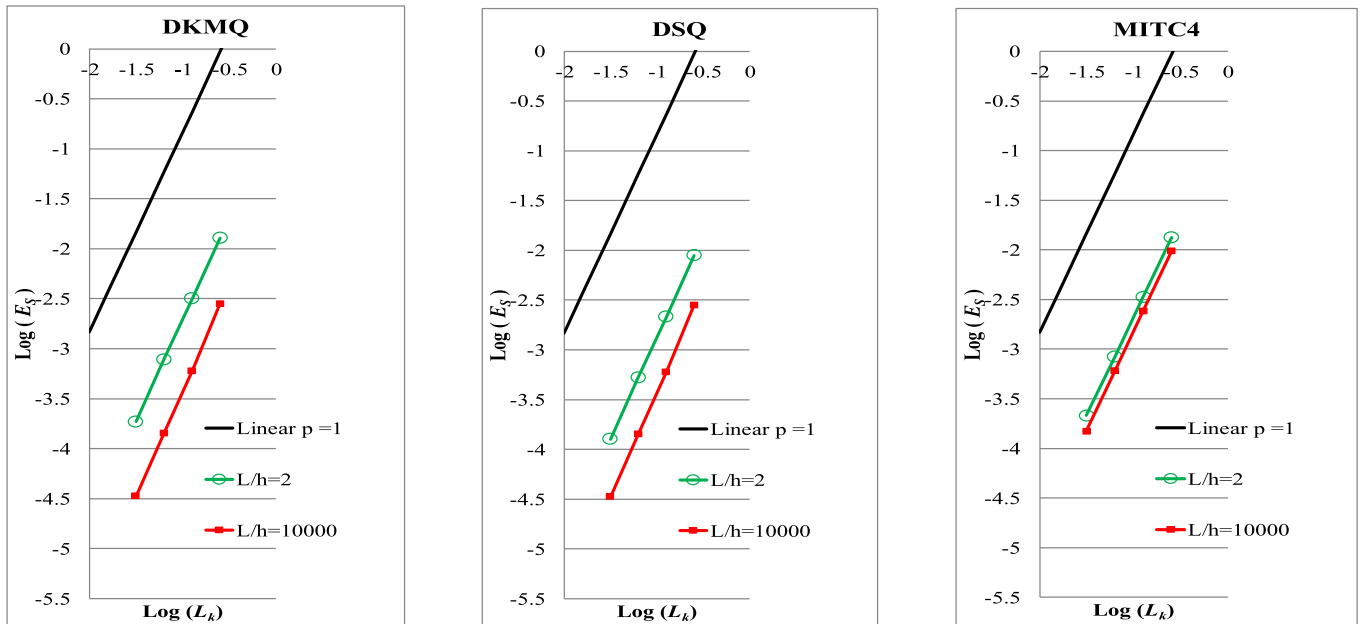
Fig. 13. Convergence behavior of clamped square plate by using s -norm with distorted mesh.Fig. 14. Simply supported square plate ($E = 1.7472 \times 10^7$; $\nu = 0.3$; $L = 1$; $f_z = -1$).

Table 4Bending and shear energy for simply supported square plate and various L/h .

L/h	$\ \Pi_{int}^b\ ^2$ (%)		$\ \Pi_{int}^s\ ^2$ (%)		$\ \Pi_{int}^{Ref}\ ^2$ (%)	
2	1.362×10^{-7}	73.07%	5.020×10^{-8}	26.93%	1.864×10^{-7}	100%
4	1.090×10^{-6}	91.56%	1.004×10^{-7}	8.44%	1.190×10^{-6}	100%
10	1.702×10^{-5}	98.55%	2.506×10^{-7}	1.45%	1.728×10^{-5}	100%
10^2	1.702×10^{-2}	99.99%	2.158×10^{-6}	0.01%	1.703×10^{-2}	100%
10^3	1.702×10^1	100%	7.592×10^{-6}	0%	1.702×10^1	100%
10^4	1.702×10^4	100%	6.937×10^{-5}	0%	1.702×10^4	100%

**Fig. 15.** Relative error for simply supported square plate with uniform mesh for $L/h = 2$ and 10^4 .**Fig. 16.** Convergence behavior of SS square plate by using s -norm with uniform mesh.

9.2. Simply supported (SS) square plate

A simply supported square plate with hard support conditions ($w = \beta_s = 0$) under uniform loading f_z is also studied (Fig. 14). The solution using DKMQ elements with $N \times N = 128 \times 128$ is considered as reference solution.

Table 4 shows the percentage of bending and shear energy for a simply supported square plate using $N \times N = 128 \times 128$ DKMQ elements for different L/h . We can see the strong influence of the boundary conditions: the percentage of bending strain energy is very important for the SS case compared to the clamped case (Table 3) even for very thick plates, (bending strain energy is 73% of the total strain energy for $L/h = 2$). Transverse shear effects quickly vanish for $L/h > 10$.

In Fig. 15 we report the results obtained using the three elements in the extreme situations of $L/h = 2$ and $L/h = 10^4$, in a classical manner, i.e. E_s versus N number of elements between A and B.

For coarse mesh ($N = 4$) the results are different with an advantage to DKMQ and DSQ, compared to MITC4, but all elements converge quickly and the relative errors E_s is again very small.

The convergence behavior of DKMQ, DSQ and elements for $L/h = 2$ and 10^4 is presented in Fig. 16 for uniform mesh and on Fig. 17 for distorted mesh for $N = 4, 8, 16$ and 32 respectively.

For the two cases analyzed, the relative error found by using DKMQ and DSQ elements is smaller than for MITC4 element. For thin plate problems, DKMQ and DSQ give the same results. We also observe in these tests that the behavior of the DKMQ element is similar to the DKQ elements when the plate is thin and similar to MITC4 element for thick plate situations.

9.3. Clamped circular plate with uniform load

We now consider a clamped circular plate with hard support conditions ($w = \beta_y = \beta_x = 0$) under uniform loading f_z (Fig. 18).

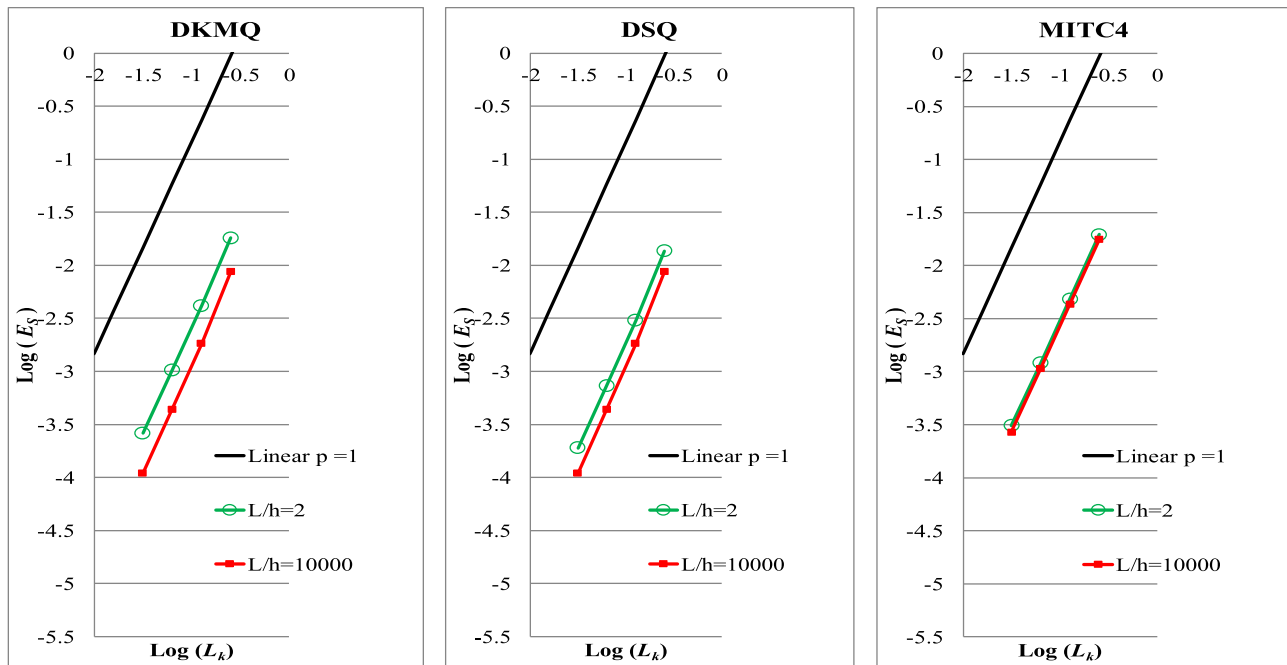


Fig. 17. Convergence behavior of SS square plate by using s -norm with distorted mesh.

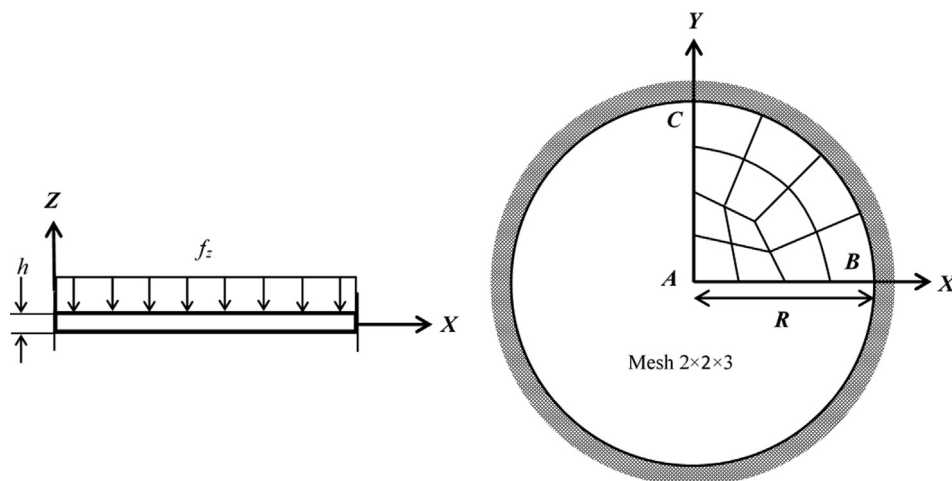


Fig. 18. Clamped circular plate ($E = 1.7472 \times 10^7$; $\nu = 0.3$; $R = 1$; $f_z = -1$).

Because of symmetry, only area ABC is evaluated. In this problem an analytical solution exists [15,18–19] and we can use the exact expression of the total strain energy to evaluate the relative energy errors. The convergence analysis of DKMQ, DSQ and MITC4 elements for $R/h=2$ and 10^4 is proposed using meshes with $N \times N \times 3 = 8 \times 8 \times 3$, $16 \times 16 \times 3$, $24 \times 24 \times 3$, $32 \times 32 \times 3$ and $64 \times 64 \times 3$ (one quarter ABC is divided in 3 zones and for each zone we consider $N \times N$ elements, see Fig. 18).

Exact energy for clamped circular plate [18–19]:

$$\Pi_{int}^{Ex} = \Pi_{int}^b + \Pi_{int}^s = \frac{f_z^2 R^6 \varphi}{384 D_b} \left(1 + \frac{4(h/R)^2}{\kappa(1-\nu)} \right)$$

φ defines the size of the sector analyzed ($\varphi = \pi/2$ for our analysis).

Table 5

Bending, shear and total exact energy for clamped circular plate and various R/h .

R/h	$\ \Pi_{int}^b\ ^2(\%)$		$\ \Pi_{int}^s\ ^2(\%)$		$\ \Pi_{int}^{Ex}\ ^2(\%)$	
2	2.045×10^{-8}	36.84%	3.506×10^{-8}	63.16%	5.552×10^{-8}	100%
4	1.636×10^{-7}	70.00%	7.012×10^{-8}	30.00%	2.337×10^{-7}	100%
10	2.557×10^{-6}	93.58%	1.753×10^{-7}	6.42%	2.732×10^{-6}	100%
10^2	2.557×10^{-3}	99.93%	1.753×10^{-6}	0.07%	2.558×10^{-3}	100%
10^3	2.557×10^0	100%	1.753×10^{-5}	0%	2.555×10^0	100%
10^4	2.557×10^3	100%	1.753×10^{-4}	0%	2.557×10^3	100%

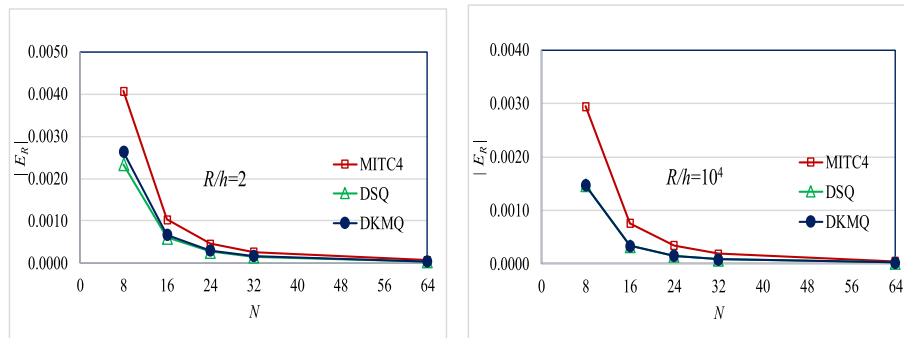


Fig. 19. Convergence of E_R for clamped circular plate using DKMQ, DSQ and MITC4 for $R/h=2$ and $R/h=10^4$.

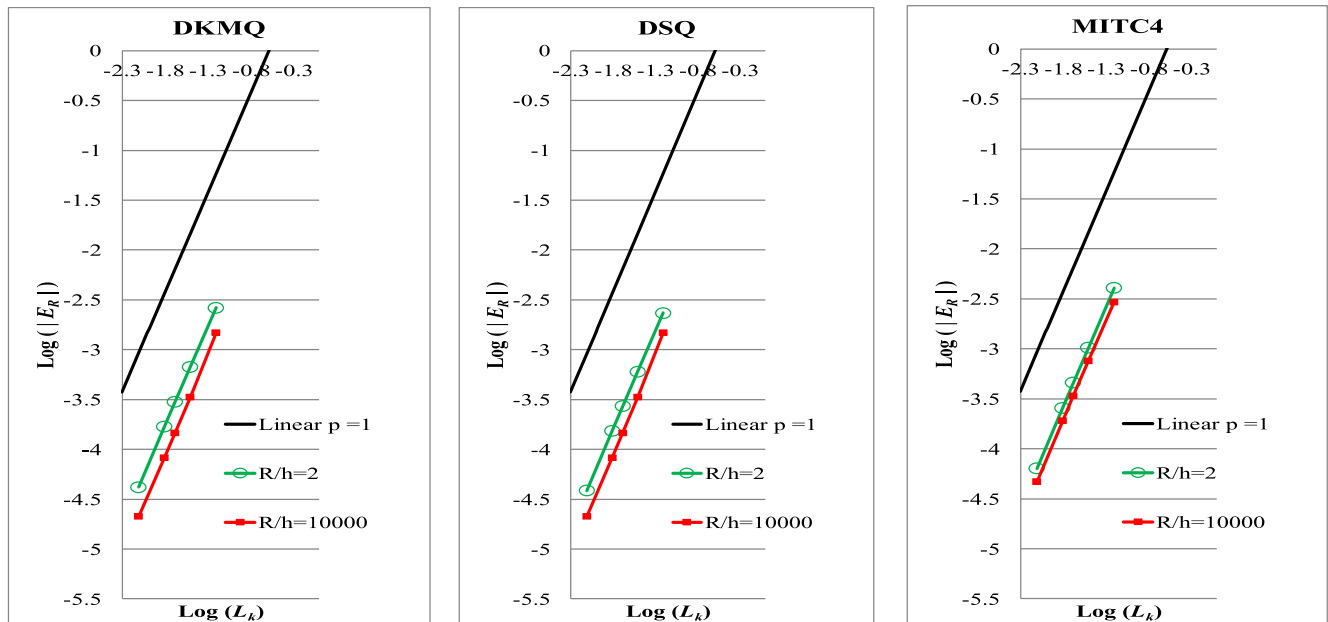


Fig. 20. Convergence behavior of clamped circular plate by using $\text{Log}(|E_R|)$ versus $\text{Log}(L_k)$.

Table 5 shows the percentage of bending and shear energy compared to the total exact strain energy for the clamped circular plate for different R/h . Results are similar compared to the clamped square plate (Table 3).

On Fig. 19 we report the results we have obtained using DKMQ, DSQ and MITC4 elements for extreme values of R/h , i.e. $R/h=2$ and 10^4 considering the energy error indicator E_R (94).

We observe a good convergence for three elements towards the exact solution with a slight advantage for DKMQ and DSQ. Three elements are converging from below (large stiffness) in the thick plate case whereas from above (smaller stiffness) in the thin plate case.

In Fig. 20, we present in the spirit of s -norm tests the results $\text{Log } E_R$ as function of $\text{Log } L_k$, with meshes $N \times N \times 3 = 8 \times 8 \times 3$,

$16 \times 16 \times 3$, $24 \times 24 \times 3$, $32 \times 32 \times 3$ and $64 \times 64 \times 3$. Again we observe the uniform linear convergence behavior for all elements.

9.4. Simply supported (SS) circular plate with uniform load

A simply supported circular plate with soft support conditions ($w = 0$) under uniform loading f_z (Fig. 21). The convergence behavior of DKMQ, DSQ and MITC4 elements for $R/h = 2$ and 10^4 is studied for different meshes.

Exact energy for simply supported (SS) circular plate [18,19]:

$$\Pi_{int}^{Ex} = \Pi_{int}^b + \Pi_{int}^s = \frac{f_z^2 R^6 \varphi}{384 D_b} \left(\frac{(7 + \nu)}{(1 + \nu)} + \frac{4(h/R)^2}{\kappa(1 - \nu)} \right)$$

φ defines the size of the sector analyzed and $\varphi = \pi/2$ for our analysis.

Table 6 shows the percentage of bending, shear energy on the total exact strain energy for a simply supported of circular plate based on exact solution for different R/h .

The percentage of bending strain energy is very important for the SS case (Table 6) compared to the clamped case (Table 5) even for very thick plates. Hence performing finite elements to represent bending energy is an advantage for the same number of *dof* for a given analysis. On Fig. 22 we report the results we obtain using DKMQ, DSQ and MITC4 for extreme values of R/h , i.e. $R/h = 2$ and 10^4 . We consider meshes with $N \times N \times 3 = 8 \times 8 \times 3$, $16 \times 16 \times 3$, $24 \times 24 \times 3$, $32 \times 32 \times 3$ and $64 \times 64 \times 3$. Both elements are rapidly converging to the exact solution for both thick and thin plates.

In Fig. 23, results using $\log E_R$ in terms of $\log L_k$, are presented for meshes $N \times N \times 3 = 8 \times 8 \times 3$, $16 \times 16 \times 3$, $24 \times 24 \times 3$,

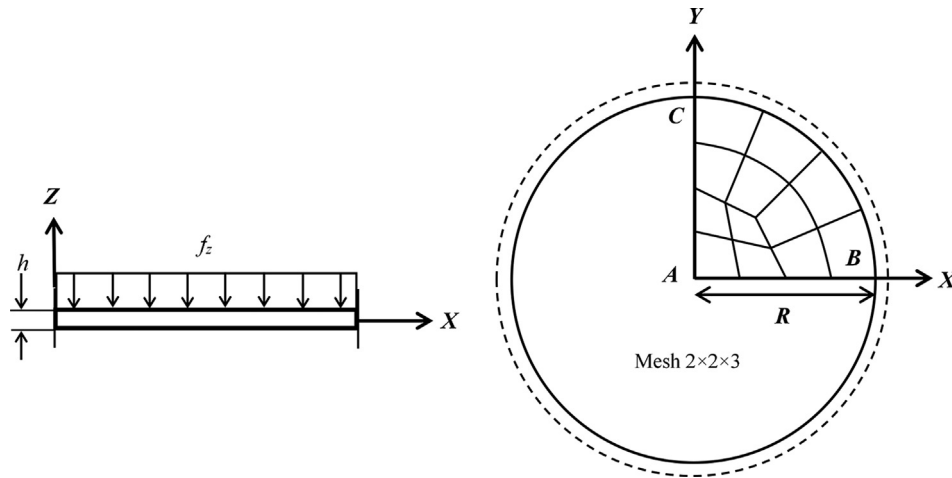


Fig. 21. Simply supported circular plate ($E = 1.7472 \times 10^7$; $\nu = 0.3$; $R = 1$; $f_z = -1$).

Table 6
Bending, shear and total exact energy for simply supported circular plate and various R/h .

R/h	$\ \Pi_{int}^b\ ^2(\%)$		$\ \Pi_{int}^s\ ^2(\%)$		$\ \Pi_{int}^{Ex}\ ^2(\%)$	
2	1.149×10^{-8}	76.61%	3.506×10^{-8}	23.39%	1.499×10^{-8}	100%
4	9.188×10^{-7}	92.91%	7.012×10^{-8}	7.09%	9.889×10^{-7}	100%
10	1.436×10^{-5}	98.79%	1.753×10^{-7}	1.21%	1.453×10^{-5}	100%
10^2	1.436×10^{-2}	99.99%	1.753×10^{-6}	0.01%	1.436×10^{-2}	100%
10^3	1.436×10^1	100%	1.753×10^{-5}	0%	1.436×10^1	100%
10^4	1.436×10^4	100%	1.753×10^{-4}	0%	1.436×10^4	100%

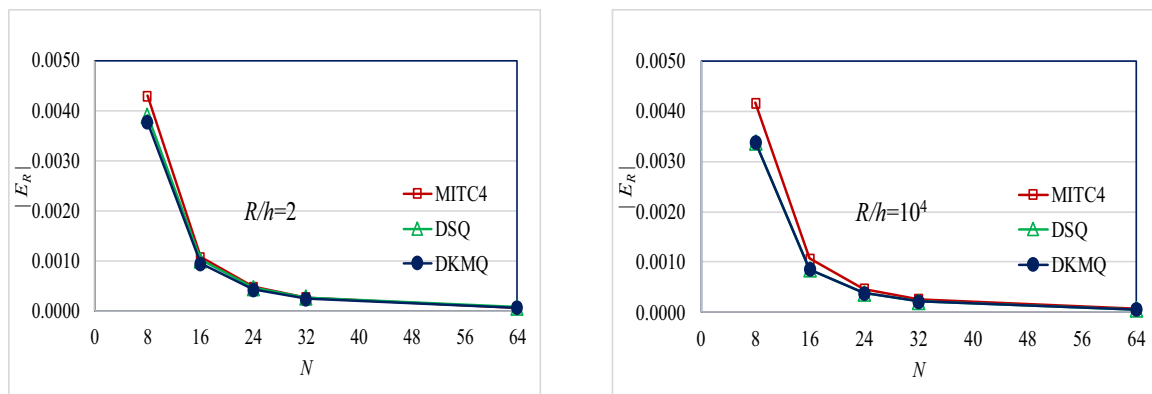


Fig. 22. Convergence of E_R for SS circular plate using DKMQ and MITC4 for $R/h = 2$ and $R/h = 10^4$.

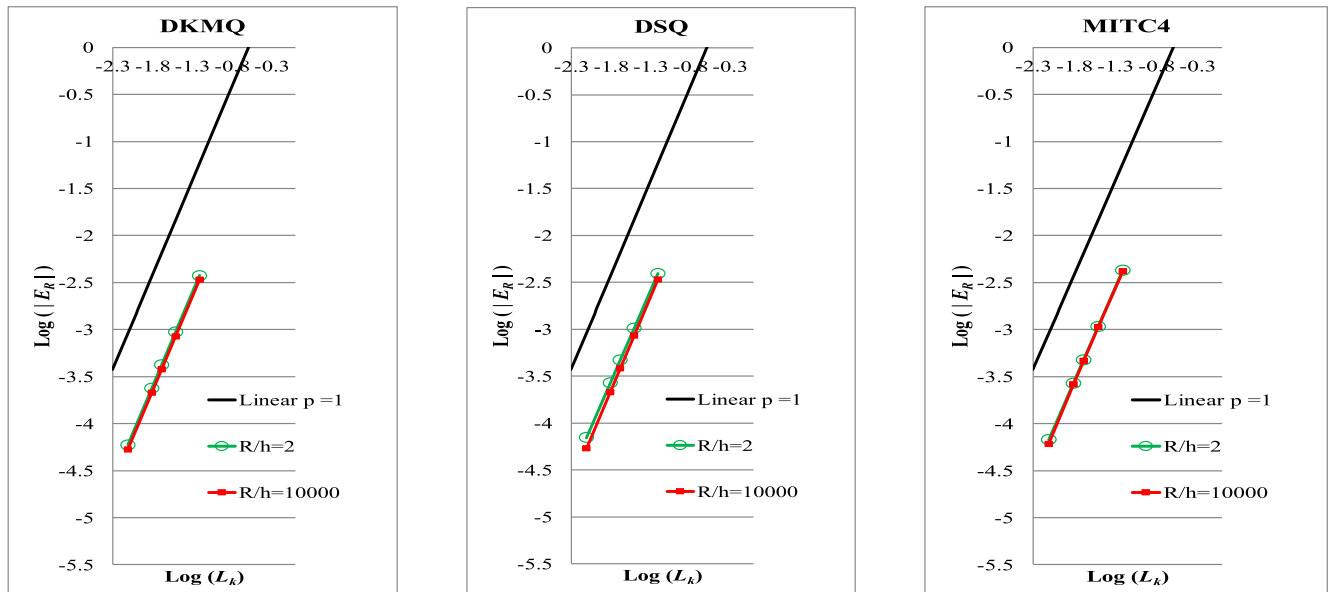


Fig. 23. Convergence behavior of SS circular plate by using $\log(|E_R|)$ versus $\log(L_k)$.

$32 \times 32 \times 3$ and $64 \times 64 \times 3$. Results one more confirm the uniform linear convergence of all elements with minor differences on the errors.

10. Concluding remarks

The paper includes detailed formulation aspects of the stiffness matrix of four quadrilateral plate bending elements with 3 *dof* per node, based on the first order Reissner-Mindlin theory. The three elements DKQ, DSQ and DKMQ have been developed between 1982 and 1993 by the senior author and former students. The fourth element MITC4 has been published in 1984, as a first element of a series of MITC plate and shell elements developed by Professor Bathe and collaborators. Even though the formulation of the four elements can be found in various documents, the present paper is written in order to understand the differences and complementarities, using a unified notation and similar assumed shear strain constraints on element sides. The paper shows that for thin plates DSQ and DKMQ should converge to DKQ without shear locking. On the other end for extremely thick plates the formulation of DKMQ is such that $\text{DKMQ} \approx \text{MITC4}$. This last result has never been shown before.

S-norm tests have been implemented and new results have been obtained and reported in this paper. The results show that uniform linear convergence is observed for the four elements even if the DKQ, DSQ and DKMQ are based on incomplete quadratic interpolation of the rotations. However, the relative error in strain energy is slightly smaller in DKQ, DSQ and DKMQ compared to MITC4 for square plates, either with uniform or distorted meshes. Due to the simplicity of formulation, the DKMQ element is much preferable than DKQ and DSQ since it is valid for thin to thick plates (not the case for DKQ) and it fully satisfies the requirement for a robust and convergent element (satisfying all types of patch tests which is not the case for DSQ).

The shear energy representation in DKMQ is also simpler than for DSQ. It is also seen that the contribution of the quadratic terms of DKMQ can be introduced without much effort to enhance MITC4 for the bending behavior of thin plates.

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